

Title- Protein Structure Prediction of Human Insulin Using SWISS-MODEL

Introduction

Protein structure prediction is a bioinformatics technique used to determine the three-dimensional (3D) structure of proteins. The SWISS-MODEL repository provides experimentally determined and homology-based protein structures that help researchers understand protein function.

Objective

To visualize and study the three-dimensional structure of Human Insulin using the SWISS-MODEL repository.

Materials and Software Used

1. UniProt Database
2. SWISS-MODEL Repository
3. Internet Browser

Methodology

1. Opened the SWISS-MODEL repository.
2. Searched for P01308 (INS_HUMAN).
3. Selected the Human Insulin protein.
4. Visualized the 3D protein structure.
5. Captured screenshots from different viewing angles.

Results and Discussion

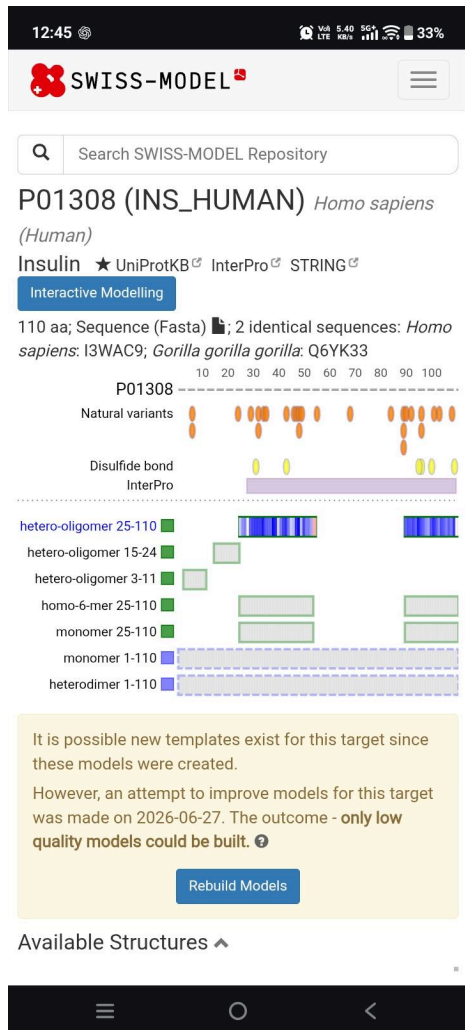
The Human Insulin protein structure was successfully visualized using SWISS-MODEL. The 3D model shows the folding pattern of the insulin protein. Different viewing angles help in understanding the spatial arrangement of the protein and its structural organization.

Structural Features

- ☐ Protein: Human Insulin (INS_HUMAN)
- ☐ UniProt ID: P01308
- ☐ Organism: Homo sapiens
- ☐ Structure Type: 3D Protein Structure
- ☐ Tool Used: SWISS-MODEL

Screenshot 1

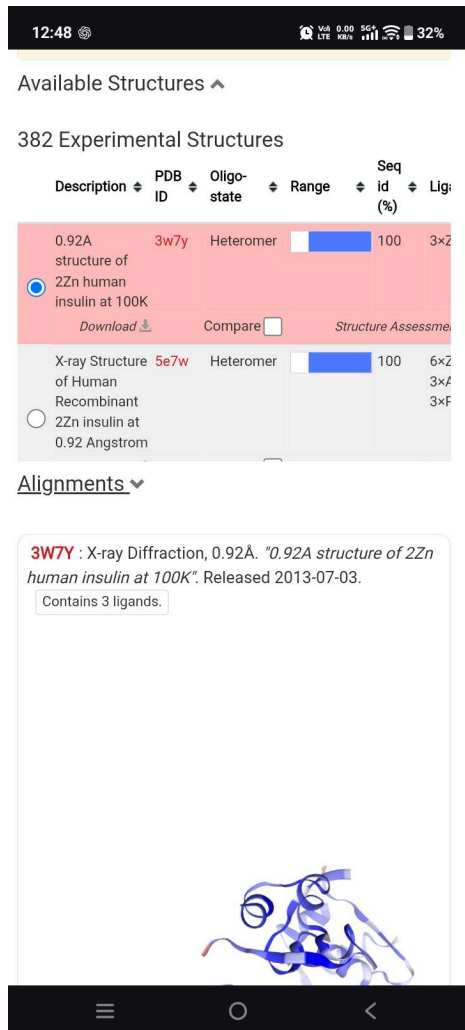
Human Insulin (P01308) Protein Information



Protein information retrieved from the SWISS-MODEL repository.

Screenshot 2

Available Structure of Human Insulin (3W7Y)



Available experimental structure of Human Insulin (PDB ID: 3W7Y) selected from the SWISS-MODEL repository for protein structure visualization.

Screenshot 3

3D Structure of Human Insulin (Front view & Rotated View)

Front view

12:53

Via 1.5G LTE 5G+ 32%

Insulin P01308 UniProtKB InterPro

Toggle Identical (BDF)

P01308 FVNQHLCGSHLVEALYLVCGERGFF 49
3w7y . (BDF) FVNQHLCGSHLVEALYLVCGERGFF 25
P01308 YTPKT 54
3w7y . (BDF) YTPKT 30

Insulin P01308 UniProtKB InterPro

Toggle Identical (ACE)

P01308 GIVEQCCTSIICSLYQLENYCN 110
3w7y . (ACE) GIVEQCCTSIICSLYQLENYCN 21

3W7Y : X-ray Diffraction, 0.92Å. "0.92A structure of 2Zn human insulin at 100K". Released 2013-07-03.

Contains 3 ligands.

⚙️

Cartoon ▾

📷 ▴

▶

⬆

↺

Schwede Group
Contact Us
Terms of use

SWISS-MODEL is part of SIB's portfolio of open databases and software tools.

⬆ Back to top

☰

○

<

Rotated view

12:54 36.9 5G+ 31%

Insulin P01308 UniProtKB InterPro

Toggle Identical (BDF)

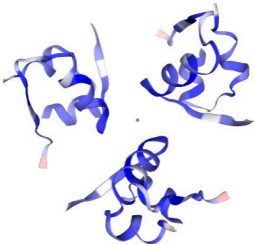
⚙ P01308 FVNQHLCGSHLVEALYLVCGERGFF 49
 3w7y. (BDF) FVNQHLCGSHLVEALYLVCGERGFF 25
 P01308 YTPKT 54
 3w7y. (BDF) YTPKT 30

Insulin P01308 UniProtKB InterPro

Toggle Identical (ACE)

⚙ P01308 GIVEQCCTSIICSLYQLENYCN 110
 3w7y. (ACE) GIVEQCCTSIICSLYQLENYCN 21

3W7Y : X-ray Diffraction, 0.92Å. "0.92Å structure of 2Zn human insulin at 100K". Released 2013-07-03.
 Contains 3 ligands.



⚙ Cartoon ▶ 📷 ▶ ▶ ⬆️ ↺

Schwede Group Contact Us Terms of use
 SWISS-MODEL is part of SIB's portfolio of open databases and software tools.
 ↑ Back to top

Rotated view of the Human Insulin 3D structure showing the protein from a different angle.

Conclusion

The 3D structure of Human Insulin was successfully visualized using SWISS-MODEL. Protein structure prediction helps understand protein function and is widely used in bioinformatics and drug discovery.

References

1. <https://swissmodel.expasy.org>
2. <https://www.uniprot.org>

